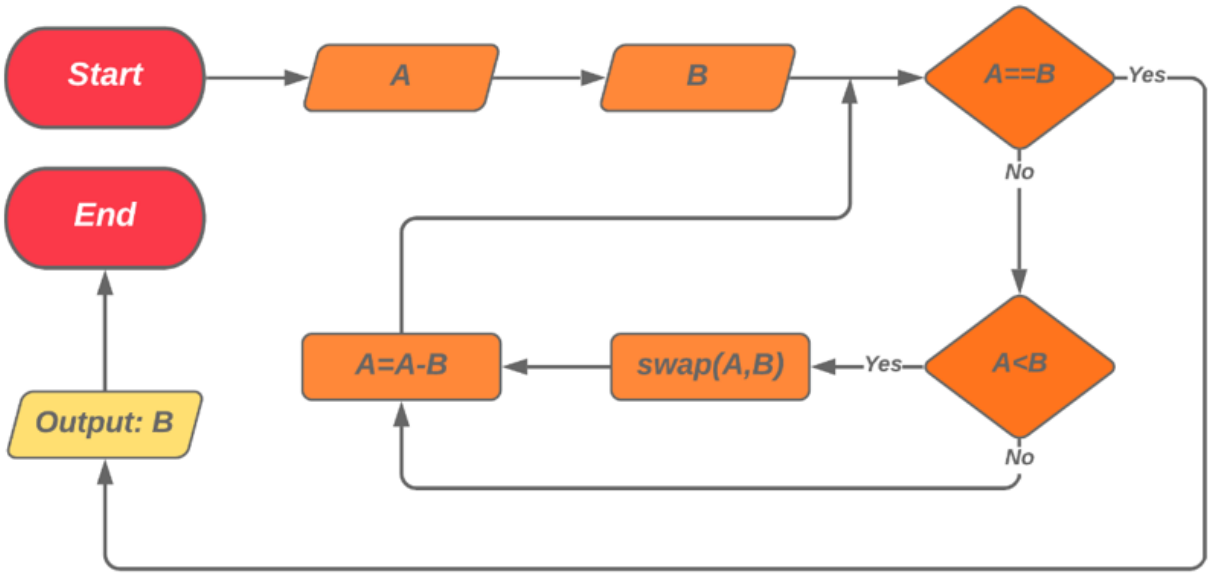


Figure 15.2 A flow chart (in this example, one for the algorithm computing the Greatest Common Divisor (GCD) of two numbers), is a special type of graph.

Besides that, we as humans often need to visualize a graph for better comprehension: consider for instance a flow chart (which, not surprisingly, is a graph!), like the one shown in figure 15.2: when we can visualize it, it's easy both to follow the flow and to get a high-level idea of its overall structure. Check out its formal definition, in terms of graph's vertices and edges, here:

```
G = (
  V = [Start, A, B, A==B, A<B, swap(A,B), A=A-B, Output: B,
End],
  E = [Start -> A, A -> B, A==B -[Yes]-> Output: B, A==B -
[No]-> Output: B, A<B -[Yes]-> swap(A,B), A<B -[No]-> A=A-B,
swap(A,B) -> A=A-B, A=A-B -> A==B, Output: B -> End]
)
```